Electrical Power – General Data

1. General

Aircraft 28 Volt DC electrical power is provided by two starter-generator systems in conjunction with one 24 volt battery (main). The DC system is a single wire type installation using the airframe structure as ground return. The main battery supplies the DC power necessary when the generators are off-line and is also used to start the engines. Aircraft with Mod 6/2315 also include a small 24 volt avionics start battery. The battery is used to supplement electrical power supply to certain avionics systems during an engine start. Further information on the avionics start battery is provided in 24-30-32, Avionics Start Battery (Mod 6/2315) – General Data.

An external 28 Volt DC power source can be connected to the aircraft for engine starting and test procedures.

DC circuits are protected by thermal push-pull circuit breakers.

The main battery is a 24 volt, 48 amp-hour lead acid type located beneath the floor of the rear baggage compartment. On aircraft incorporating S.O.O. 6371, the main battery has been relocated to the nose baggage compartment. The main battery supplies power to the electrical system when the MASTER switch is selected to ON, and the POWER SOURCE switch is selected to BATTERY (Pre Mod 6/2314). When Mod 6/2314 is incorporated, the main battery supplies power to the electrical system when the BATTERY switch is selected to ON. The main battery also supplies power independently through a hot battery bus to the entrance lights, forward and rear baggage compartment lights, and the cockpit dome light and clock.

2. Removal/Installation

A. Safety Precautions

The following safety precautions should be observed prior to the removal or installation of any electrical equipment or components in the aircraft.

1) Refer to appropriate wiring diagram in the Wiring Diagram Manual, PSM 1-64-2W, prior to removal and installation of any electrical equipment or components and when disconnecting or connecting electrical wiring.

2) Ensure that MASTER and POWER SOURCE switches are set to OFF (Pre Mod 6/2314); or BATTERY, EXTERNAL and GENERATOR switches are set to OFF (Post Mod 6/2314).

3) Disconnect external power source and disconnect main battery. Pull out applicable circuit breaker and insert clip to prevent push-to-reset.

4) If units removed are not to be immediately replaced, ensure that connector ends are capped and stowed, and all loose wires are insulated and stowed.

5) After installation, remove clip and push in circuit breaker; do a function test on unit.

Removal and installation procedures for the electrical equipment is self-evident, except for the main battery (refer to 24-30-31, Main Battery Power – Maintenance Practices). Equipment grounding and bonding surfaces must be cleaned before and after installation, any bared surfaces must be refinished. When test or repair requires removal of cables from terminals, the terminal nuts must be tightened to the torque values given in Table 1 — Wrench Torque Values for Securing Terminal Nuts when cables are reconnected.

Table 1 Wrench Torque Values for Securing Terminal Nuts

TERMINAL STUD SIZE	TERMINAL LUG METAL	TORQUE-VALUE (POUND-INCHES)
1/2 inch	Copper	140 – 155
3/8 inch	Copper	110 – 125
3/8 inch (Gen Shunt Ground)	Copper	110 – 124
5/16 inch	Copper	80 – 85
AN3	Steel	40 – 45
1/4 inch	Copper	40 – 45
1/4 (Bat. Shunt Ground)	Copper	40 – 45
No. 10	Copper	15 – 20
No. 8	Copper	9 – 12
No. 6	Copper	Tighten to depress lockwasher